

Arithmetic sequences



1

a No

b Yes

c No

d Yes

e No

f No

g Yes

h No

2 By choosing any term after the first and subtracting the preceding term from it.

3

a $d = -2$

b $d = 11$

c $d = -50$

d $d = \frac{3}{7}$

4

a 16, 20,

b -10, -14

c 24, 29

d 8, 9.5

5

a 12, 24

b -5, 5, 15

6

a -10, -6, -2, 2

b -8, -10, -12, -14

c $a, a + d, a + 2d, a + 3d$

7

a $d = 0.9$

b $n = 11$

8 -6, 0, 6, 12, 18

9 $x = -7$

10

a $d = 4$

b $u_1 = 5$

c $u_{11} = 45$

11

a $d = -3$

b $u_1 = 4$

c -35

12 $d = -12$

13

a $n \leq 5$

b 5

c 6

14

a $d = \frac{y-x}{4}$

b $u_1 = \frac{9x-5y}{4}$



c $\frac{-3x+5y}{2}$

15

a $u_1 = 0$

b $u_9 = -16$

16 $n = 27$

17 $n = 117$

Explicit rules

18

a i $d = 7$ ii $u_n = 7n - 2$ iii $u_{10} = 68$

b i $d = -0.8$ ii $u_n = -0.8n + 17.8$
iii $u_{10} = 9.8$

c i $d = -7$ ii $u_n = -7n + 17$
iii $u_{10} = -53$

d i $d = \frac{3}{4}$ ii $u_n = \frac{3n+17}{4}$
iii $u_{10} = \frac{47}{4}$

19

a i 11, 14, 17, 20 ii 3 b i 11, 21, 31, 41 ii 10

c i -7, -10, -13, -16 ii -3

20

a i $u_1 = 15$ ii $d = 5$ iii $u_5 = 35$

b i $u_1 = 20$ ii $d = -8$ iii $u_5 = -12$

c

d

i $u_1 = 3$

ii $d = 12$



iii $u_5 = 51$

21

a 6

b 8

c $u_n = 6n + 2$

d 152

Applications

22 \$2670

23 1510 cm

24

a 50 m

b 200 m

c 500 m

25

a $d = 35$

b $u_1 = 151$

c $n = 22$

26

a \$1 800

b $V_n = 1\,800 - 50n$

c $n = 18$